

Project Report Outline

1. Title Page [1 page]

- Project Title
- Course Name: Integrative Data Analytics
- Team Members
- Submission Date
- Instructor's Name

2. Introduction [1-2 pages]

- **Background and Context** – Explain the business or real-world context of the dataset/project
- **Objectives of the Project**- State the goals of analysis

3. Methodology [1-2 pages]

- **Dataset Description** – Source, size, variables, type of data
- **Analytical Techniques** – Detail preprocessing, EDA, PCA, and any optional models used
- **Tools and Technologies** – Mention platforms (e.g., Python, Jupyter) and key libraries

4. Analysis and Results [4-6 pages]

- **Data Preprocessing**
 - Explain every step taken (handling missing values, encoding, outliers, etc.)
- **Exploratory Data Analysis (EDA)**
 - Discuss statistical summaries and key visualizations
 - Include insights from patterns, distributions, correlations
- **Principal Component Analysis (PCA)**
 - Describe the purpose of PCA in your project
 - Include scree plots and interpretation of components

5. Discussions [1 page]

- **Interpretation of Results**
 - Connect analysis results to objectives
 - Highlight major findings and their implications

References

Rubric for Capstone Report (100 points)

Section	Points	Criteria
Title Page	5 pts	Complete and properly formatted
Introduction	15 pts	Clear background and objectives
Methodology	15 pts	Dataset described well, techniques and tools explained clearly
Data Preprocessing	15 pts	Logical, well-documented steps with justifications
Exploratory Data Analysis (EDA)	15 pts	Appropriate visualizations, meaningful insights, good interpretation
Principal Component Analysis	15 pts	Steps clearly explained, graphs included, interpretation provided
Discussion	15 pts	Results interpreted in line with problem and objectives
Clarity and Presentation	5 pts	Well-organized, free from major grammar/spelling issues
Professionalism & Completeness	5 pts	Report is polished, complete, and demonstrates effort